Information Processing

Formulae

Standard Laplace Transforms

Signal	Transform	ROC
$u(t)$	$\frac{1}{s}$	$\mathrm{Re}(s) > 0$
$tu(t)$	$\frac{1}{s^2}$	$\mathrm{Re}(s) > 0$
$\delta(t-\tau),\ \tau \geq 0$	$e^{-s\tau}$	$\forall s$
$e^{-at}u(t)$	$\frac{1}{s+a}$	$\mathrm{Re}(s) > -a$
$te^{-at}u(t)$	$\frac{1}{(s+a)^2}$	$\mathrm{Re}(s) > -a$
$\cos(w_1 t)u(t)$	$\frac{s}{s^2+w_1^2}$	$\mathrm{Re}(s) > 0$
$\sin(w_1 t)u(t)$	$\frac{w_1}{s^2+w_1^2}$	$\mathrm{Re}(s) > 0$
$e^{-at}\cos(w_1 t)u(t)$	$\frac{s+a}{(s+a)^2+w_1^2}$	$\mathrm{Re}(s) > -a$
$e^{-at}\sin(w_1 t)u(t)$	$\frac{w_1}{(s+a)^2+w_1^2}$	$\mathrm{Re}(s) > -a$

1. This question caries 40% of the marks.

a) For each of the following continuous-time signals, state whether they are causal or noncausal, periodic or nonperiodic, even or odd symmetric. If periodic, state the period.

i) $x_1(t) = |2\cos(3\pi t)|$ [2]

Solution:
noncausal, periodic, even
period $T = 1/3$

ii) $x(t) = \sin 2t + \sin 2\pi t$ [2]

Solution:
noncausal, nonperiodic, odd

b) Consider the signal

$$x(t) = \begin{cases} 1-t & 0 \le t \le 1 \\ 0 & \text{otherwise} \end{cases}$$

For each of the following signals $y(t)$, find an expression for $y(t)$ not involving x. Describe, using illustrative sketches, the sequence of steps that must be taken to form $y(t)$ from $x(t)$.

i) $y(t) = x(-2t-1)$ [3]

Solution:

$$y(t) = x(-2t-1) = \begin{cases} 1-[-2t-1] = 2t+2 & \text{for } -1 \le t \le -0.5 \\ 0 & \text{otherwise} \end{cases}$$

$y(t)$ is formed from $x(t)$ by:

- $t \leftarrow (t-1)$: shift in the direction of +ve time by 1
- $t \leftarrow (2t)$: compress on time axis by factor of 2
- $t \leftarrow (-t)$: reflection in $t = 0$

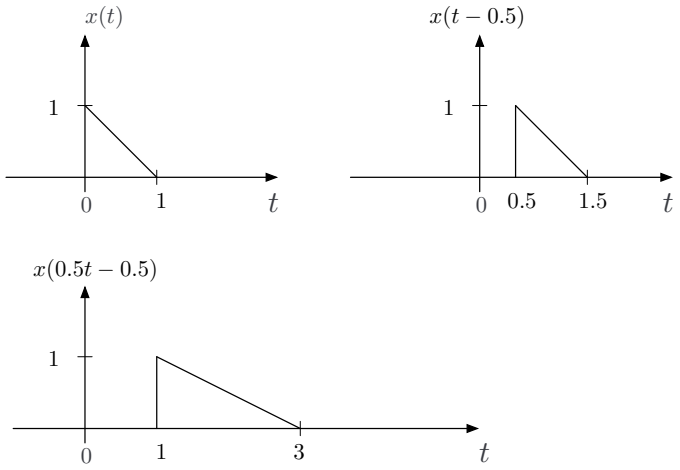
ii) $y(t) = x(0.5t - 0.5)$ [3]

Solution:

$$y(t) = x(0.5t - 0.5) = \begin{cases} 1 - [0.5t - 0.5] = -0.5t + 1.5 & \text{for } 1 \le t \le 3 \\ 0 & \text{otherwise} \end{cases}$$

$y(t)$ is formed from $x(t)$ by:

- $t \leftarrow (t - 0.5)$: shift in the direction of +ve time by 0.5
- $t \leftarrow (0.5t)$: expand on time axis by factor of 2



c) An LTI system $H(z)$ has impulse response $h[n] = 0.1^n u[n]$.

i) Derive and simplify an expression for the step response of $H(z)$, denoted $s[n]$. [3]

Solution:

$$\begin{aligned} & h[n] = 0.1^n u[n] \\ \Longrightarrow & s[n] = \sum_{k=-\infty}^{n} h[k] = \sum_{k=0}^{n} 0.1^n \\ & = \frac{1 - r^{n+1}}{1 - r} = (1 - 0.1^{n+1})/0.9. \end{aligned}$$

ii) Show how $h[n]$ can be obtained from your expression for $s[n]$. [3]

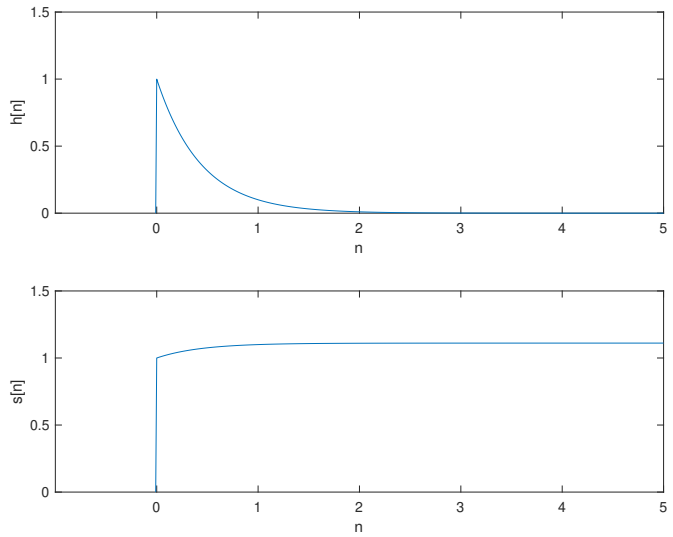
Solution:

$$h[n] = s[n] - s[n-1] = \Big[(1 - 0.1^{n+1}) - (1 - 0.1^n)\Big]/0.9$$
$$= \Big[(-0.1)(-0.1)^n + (0.1)^n\Big]/0.9 = 0.1^n(1 - 0.1)/0.9 = 0.1^n$$

Since $s[n] = 0$ for $n < 0$, then $h[n]$ can be written $h[n] = 0.1^n u[n]$.

iii) Draw illustrative labelled sketches of $h[n]$ and $s[n]$. [2]

Solution:



d) Consider a signal $x_1(t)$ with Fourier transform

$$X_1(j\omega) = \begin{cases} 2 & |\omega| < \omega_1 \\ 0 & \text{otherwise} \end{cases}$$

i) Find the signal $x_1(t)$ in terms of ω_1. Show your working. [3]

Solution:

$$x_1(t) = \frac{1}{2\pi}\int_{-\omega_1}^{\omega_1} 2e^{j\omega t}d\omega = \frac{2\sin\omega_1 t}{\pi t}$$

ii) Draw labelled sketches of $x_1(t)$ and $X_1(j\omega)$. [4]

Solution:
See part (iii) solution.

iii) Now consider a signal $x_2(t)$ with Fourier transform

$$X_2(j\omega) = \begin{cases} 2 & |\omega| < \omega_2 \\ 0 & \text{otherwise} \end{cases}$$

and $\omega_2 = \pi\omega_1$.

Draw a labelled sketch of $x_2(t)$ and state the similarities and differences between $x_1(t)$ and $x_2(t)$. [2]

Solution:
Sketches are required showing the rectangular functions in ω and corresponding sinc functions in time with relevant labelling of the axes, peak values and zero crossings.
The signals $x_1(t)$ and $x_2(t)$ are similar in that they have the shape of the well-known sinc function. They differ in that: the first zero-crossing for $x_1(t)$ is at π/ω_1 whereas for $x_2(t)$ it is $1/\pi.(\pi/\omega_1) = 1/\omega_1$, notably closer to the origin than for $x_1(t)$; periodic fluctuations of the sincs are corresponding scaled; the peak amplitude of $x_1(t)$ is $2\omega_1/\pi$ and for $x_2(t)$ this is increased by a factor π.

e) Consider the signal $x(t)$ shown in Fig. 1.1.

i) Write an expression for the signal $x(t)$. [1]

ii) Derive its Laplace transform $X(s)$ and show your working. [3]

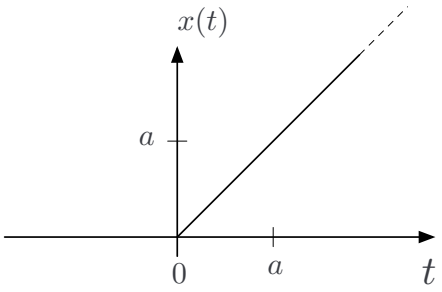


Figure 1.1

Solution: Using the differentiation property of the Laplace transform

$$-tx(t) \xleftrightarrow{\mathscr{L}u} \frac{d}{ds}X(s)$$

and noting that the Laplace transform of $u(t)$ is $\frac{1}{s}$ with ROC $\mathrm{Re}(s) > 0$

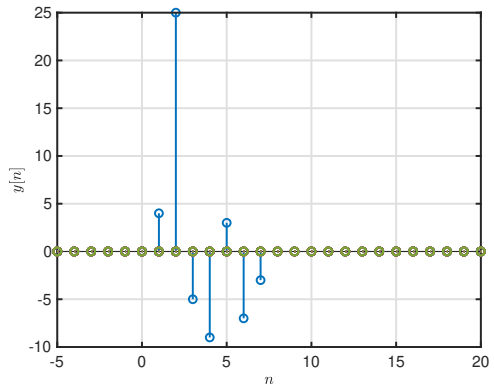
$$x(t) = tu(t)$$
$$X(s) = -\frac{dU(s)}{ds} = -\frac{d}{ds}\frac{1}{s} = \frac{1}{s^2}.$$

f) Consider the finite duration discrete-time signals $x_1[n]$ and $x_2[n]$. The nonzero values of these signals are given in Table 1. Find and sketch the signal $y[n] = x_1[n] * x_2[n]$ for the range $-5 \leq n \leq 20$ where $*$ indicates convolution. [4]

n	-3	-2	-1	0	1	2	3	4	5	6	7
$x_1[n]$		4	1	-3	-1						
$x_2[n]$							1	6	-2	3	

Table 1

Solution:



g) Consider a discrete-time signal $x[n]$ with DTFT

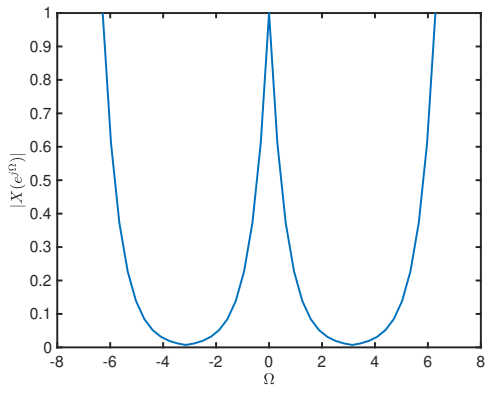
$$X(e^{j\Omega}) = \begin{cases} e^{-\pi\Omega/2} & 0 \le \Omega \le \pi \\ e^{\pi\Omega/2} & -\pi < \Omega < 0. \end{cases}$$

Draw a sketch showing the magnitude spectrum of

$$X(e^{j\Omega}), \quad -2\pi \le \Omega \le 2\pi.$$

[2]

Solution:



h) The discrete-time signal $x[n]$ with sample index n has z-transform $X(z)$ with *Region of Convergence* R_x. Showing each step of your working, prove that

$$x[-n] \xleftrightarrow{z} X\left(\frac{1}{z}\right)$$

and state the *Region of Convergence* for this z-transform. [3]

Solution:
Take z-transform $\sum_{n=-\infty}^{\infty} x[-n]z^{-n}$ and replace the variable n with $-m$ giving $\sum_{m=-\infty}^{\infty} x[m](z^{-1})^{-m}$. This is the z-transform with z replaced by z^{-1}. Accordingly the z-transform of $x[-n]$ is $X(z^{-1})$ and the region of convergence is the corresponding range of z^{-1}, giving the ROC as $\frac{1}{R_x}$.

2. This question caries 30% of the marks.

a) Summarize the similarities, differences and relationships between the Fourier transform and the bilateral Laplace transform in the context of continuous-time signals and systems. Include any relevant illustrative sketches. [7]

Solution:
Considering LTI systems, starting with the Laplace transform equation and substituting $s = \sigma + j\omega$ gives

$$H(s) = \int_{-\infty}^{\infty} h(t)e^{-st}dt$$
$$H(\sigma + j\omega) = \int_{-\infty}^{\infty} h(t)e^{-(\sigma+j\omega)t}dt = \int_{-\infty}^{\infty} \left[h(t)e^{-\sigma t}\right] e^{-j\omega t}dt$$

This expression is recognized as the Fourier transform applied to the term in square brackets so that the Laplace transform of any function of continuous time $h(t)$ is equivalent to the Fourier transform of $h(t)e^{-\sigma t}$

The Fourier representation $H(j\omega)$ is a particular case of the Laplace representation $H(s)$ for $\sigma = 0$

$$H(j\omega) = H(s)\Big|_{\sigma=0}$$

if the Fourier transform exists, so that the Fourier transform is obtained from the Laplace transform with $s = j\omega$ so that the Fourier transform is the Laplace transform but evaluated only on the $j\omega$ axis.

If it is known that the Fourier transform of a signal exists then the ROC of the Laplace transform must include the $j\omega$ axis.

An informative labelled sketch of the s-plane should be given.

b) Consider the continuous-time LTI system $H(s)$ with input signal $x(t)$, output signal $y(t)$, and impulse response

$$h(t) = ae^{-4t} - be^{-3t}.$$

It is known that

$$y(t) = \frac{19}{20}e^{t} \quad \text{for } x(t) = e^{t}$$
$$y(t) = \frac{4}{5}e^{2t} \quad \text{for } x(t) = e^{2t}.$$

i) Write an expression for $H(s)$ and find the numerical values of the constants a and b. [5]

ii) Sketch the representation of $H(s)$ on the s-plane. [1]

iii) State the d.c. gain of this system. [1]

Solution:

(i) From the impulse response given

$$h(t) = ae^{-4t} - be^{-3t} \xleftrightarrow{\mathscr{L}} H(s) = \frac{a}{s+4} - \frac{b}{s+3} \quad \text{Re}(s) > -3.$$

The eigenfunction Laplace transform property of e^{st} shows that

$$y(t) = H(s)e^{st} \quad \text{for } x(t) = e^{st}.$$

Hence, for $x(t) = e^{t}$ we obtain

$$H(1) = \frac{a}{5} - \frac{b}{4} = \frac{19}{20}$$

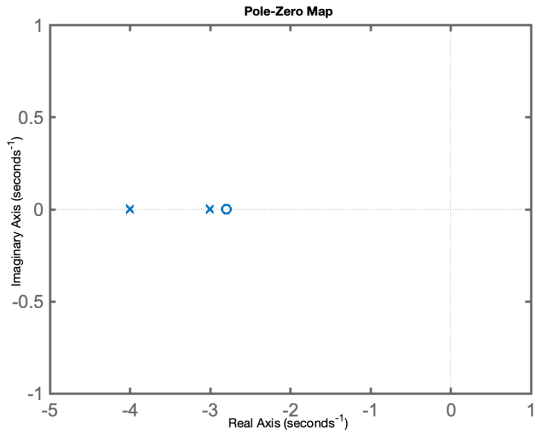
and for $x(t) = e^{2t}$ we obtain

$$H(2) = \frac{a}{6} - \frac{b}{5} = \frac{4}{5}.$$

Solving gives $a = 6$ and $b = 1$ so that

$$H(s) = \frac{6}{s+4} - \frac{1}{s+3} = 5.\frac{s+14/5}{(s+4)(s+3)}$$

(ii)



(iii) Gain at d.c. is given by $|H(0)| = \frac{6}{4} - \frac{1}{3} = 7/6$.

c) Using partial fraction expansion, find the inverse bilateral Laplace transform of the system described by

$$H(s) = \frac{s}{(s+3)(s^2+4s+5)} \qquad \text{ROC: Re}(s) > -2.$$

[8]

Solution:
PFE:

$$H(s) = \frac{A}{s+3} + \frac{Bs+C}{s^2+4s+5} = \frac{s^2(A+B)+s(4A+3B+C)+5A+3C}{(s+3)(s^2+4s+5)}$$

By comparing powers of s in the numerator, this leads to

$$\begin{aligned} A+B=0 \quad 4A+3B+C=1 \quad 5A+3C=0 \\ \Longrightarrow A=-3/2 \quad B=3/2 \quad C=5/2 \end{aligned}$$

so that

$$H(s) = -\frac{3}{2}.\frac{1}{s+3} + \frac{3}{2}.\frac{s+5/3}{s^2+4s+5}$$

Next considering standard Laplace transform pairs. For the first term, noting that the ROC of $H(s)$ is to the right of the pole at $s=-3$, use the right-sided expression

$$\begin{aligned} \frac{1}{s+a} &\xleftrightarrow{\mathscr{L}} e^{-at}u(t) \quad \mathrm{Re}(s) > -a \\ \text{so that } -\frac{3}{2}.\frac{1}{s+3} &\xleftrightarrow{\mathscr{L}} -\frac{3}{2}e^{-3t}u(t) \quad \mathrm{Re}(s) > -3 \end{aligned}$$

For the second term, noting that the ROC of $H(s)$ is to the right of the poles at $s=-2\pm j$, use

$$\begin{aligned} \frac{s+a}{(s+a)^2+w_1^2} &\xleftrightarrow{\mathscr{L}} e^{-at}\cos(w_1 t)u(t) \quad \mathrm{Re}(s) > -a \\ \frac{w_1}{(s+a)^2+w_1^2} &\xleftrightarrow{\mathscr{L}} e^{-at}\sin(w_1 t)u(t) \quad \mathrm{Re}(s) > -a \end{aligned}$$

and write

$$\begin{aligned} \frac{s+5/3}{(s+2)^2+1} &= \frac{s+2}{(s+2)^2+1} - \frac{1}{3}.\frac{1}{(s+2)^2+1} \\ &\xleftrightarrow{\mathscr{L}} e^{-2t}\cos(t)u(t) - \frac{1}{3}e^{-2t}\sin(t)u(t). \end{aligned}$$

Finally for both terms

$$\begin{aligned} H(s) &\xleftrightarrow{\mathscr{L}} -\frac{3}{2}e^{-3t}u(t) + \frac{3}{2}\left(e^{-2t}\cos(t)u(t) - \frac{1}{3}e^{-2t}\sin(t)u(t)\right) \\ &= \frac{3}{2}u(t)\left[-e^{-3t} + e^{-2t}\left(\cos(t) - \frac{1}{3}\sin(t)\right)\right] \end{aligned}$$

d) A continuous-time LTI system is described by

$$\frac{d^2y(t)}{dt^2} - 4\frac{dy(t)}{dt} + 3y(t) = \frac{dx(t)}{dt} + x(t)$$

where $x(t)$ is the input signal and $y(t)$ is the output signal.

i) Find the corresponding system function, denoted $H(s)$, as the ratio of two polynomials. [3]

ii) Find the impulse response of this system, denoted $h_s(t)$, if the system is stable. [3]

iii) Find the impulse response of this system, denoted $h_c(t)$, if the system is causal. [2]

Solution:

$$Y(s)(s^2-4s+3)=X(s)(s+1)\Longrightarrow H(s)=\frac{s+1}{s^2-4s+3}$$

$$H(s)=\frac{s+1}{s^2-4s+3}=\frac{-1}{s-1}+\frac{2}{s-3}$$

For a stable system, the ROC must include the $j\omega$ axis and so is to the left of the poles in both terms, giving $h_s(t)$ as the sum of two left-sided components

$$h_s(t)=e^t u(-t)-2e^{3t}u(-t).$$

For a causal system comprising only right-sided components

$$h_c(t)=-e^t u(t)+2e^{3t}u(t).$$

3. This question caries 30% of the marks.

a) Consider two signals

$$x_1[n] = 0.9^n u[n] \quad x_2[n] = 0.2^n u[n]$$

where $u[n]$ is the unit step function. Now consider another signal

$$y[n] = x_1[-n-2] * x_2[n-3]$$

where $*$ denotes convolution. Find the z-transform of $y[n]$. [6]

Solution:

$$x_1[n] = 0.9^n u[n] \xleftrightarrow{z} X_1(z) = \frac{1}{1-0.9z^{-1}} \quad |z| > 0.9$$
$$x_2[n] = 0.2^n u[n] \xleftrightarrow{z} X_2(z) = \frac{1}{1-0.2z^{-1}} \quad |z| > 0.2$$

$$x_1[-n-2] \xleftrightarrow{z} z^2 X_1(z^{-1}) = \frac{z^2}{1-0.9z}$$
$$x_2[n-3] \xleftrightarrow{z} z^{-3} X_2(z) = \frac{z^{-3}}{1-0.2z^{-1}}$$

$$y[n] = x_1[-n-2] * x_2[n-3] \xleftrightarrow{z} Y(z) = \frac{z^2}{1-0.9z} \cdot \frac{z^{-3}}{1-0.2z^{-1}}$$
$$= \frac{-1}{0.9z^2 - 1.18z + 0.2}.$$

b) Consider a set of sound recordings of bird song. The recordings were made on a portable digital recording device with an analogue-to-digital converter operating at a fixed sampling frequency of 48 kHz with 16 bits per sample. The device has a built-in microphone with an associated microphone amplifier with variable gain.

The bits representing each sample are denoted $Q0$, $Q1$, ..., $Q15$. Bit $Q15$ is the sign bit so that $Q15 = 0$ for positive samples and $Q15 = 1$ for negative samples. The bits $Q0$, $Q1$, ..., $Q14$ represent the unsigned magnitude of the samples where $Q0$ denotes the Least Significant Bit and $Q14$ denotes the Most Significant Bit.

A study was made of the recordings. The bit values corresponding to the samples of the signals were examined separately for the periods of time when the birds were singing (denoted SONG) and for the periods of time when the birds were briefly silent between songs (denoted PAUSE). The study also included a spectral analysis of the recordings.

The conclusions of the study were as follows.

- Bits $Q13$ and $Q14$ are always zero in both SONG and PAUSE.
- Bits $Q0 \ldots Q4$ vary with uniform distribution in PAUSE.
- There is no significant energy above 11.5 kHz.

i) State the properties of the recordings that can be inferred from the study's conclusions. [3]

ii) State what signal processing could be advantageously applied to the recorded bird song signals and explain what advantages or disadvantages such processing would have. [2]

iii) State any recommendations you would give for improving the recording process without changing the recording device, and explain the nature of the expected benefits. [2]

Solution:
(i) The signals are bandlimited to 11.5 kHz.
Since bits 13 and 14 are always zero, the recorded signal amplitude is bounded to 14 bits, corresponding to 0.25 of the full-scale dynamic range (sometimes written -12 dBFS).
The background noise level is of magnitude corresponding to 5 bits for each of the positive and negative samples. This gives a uniform PDF random signal in the range -2^5 up to $+2^5$ corresponding to $1/1000^{\text{th}}$ of the full-scale dynamic range (sometimes written -60 dBFS).

(ii) The signals can be downsampled to $f_s > 23$ kHz with the advantages of reduced data storage requirements and faster processing. There are no immediate disadvantages discussed within the scope of this course.

(iii) The microphone amplifier should be more carefully adjusted to use the maximum dynamic range but avoid clipping. If the majority of the noise is acoustic, this would not improve the SNR but would make better use of the signal dynamic range of the ADC.

c) A 60 Hz continuous-time cosinusoidal signal is sampled using a sampling frequency of 500 Hz. A total of 64 samples are acquired and stored as the discrete-time signal $x[n]$. A 64-point FFT is then applied to $x[n]$ to obtain the discrete spectrum $X[k]$ for $k = 0, 1, \ldots, 63$. Determine the value (or values) of k for which $X[k]$ has the greatest magnitude. [6]

Solution:
The frequency resolution will be given by $fs/N = 500/64 = 7.8$ Hz. The max magnitude will occur in $X[k]$ for k corresponding to frequency closest to 60 Hz. For $k = 7$ this corresponds to 54.7 Hz and for $k = 8$ this corresponds to 62.5 Hz. Hence $k = 8$ is closest and will give the $X[k]$ with greatest magnitude. By periodicity of the FFT, $X[k]$ for $k = 64 - 8$ will have the same value and therefore also have the greatest magnitude.

d) Consider a discrete-time LTI system with impulse response $h[n]$ and system function $H(z)$. The pole-zero diagram for $H(z)$ is shown in Fig. 3.1.

i) Find expressions for $h[n]$ and $H(z)$. [2]

ii) Draw a block diagram of the implementation of the system function $H(z)$, denoting the input signal $x[n]$ and the output signal $y[n]$. [2]

iii) Draw a labelled sketch of the magnitude frequency response of $H(z)$ for the range of normalized angular frequency $0 \leq \Omega < \pi$.

[3]

iv) Determine the output, $y[n]$, when the input is

$$x[n] = 3.7 + 4.4\cos(n\pi/2) - 2\cos(n\pi/4).$$

[4]

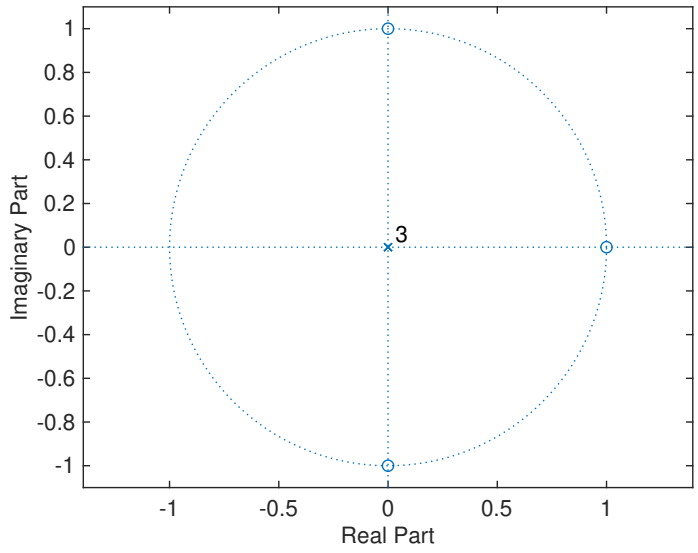
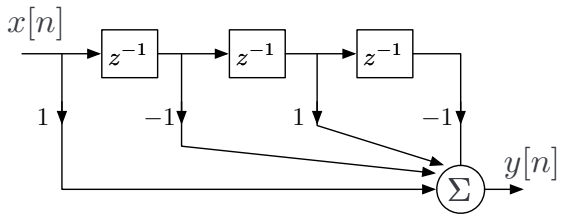


Figure 3.1

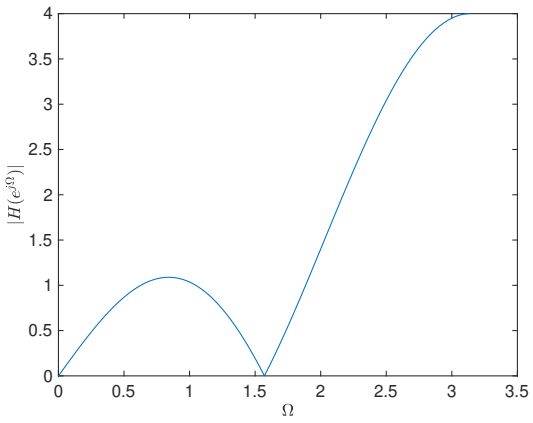
Solution:
(i) The plot shows zeros at $z = 1,\ \pm j$ and 3 poles at the origin, hence

$$H(z) = \frac{(z-1)(z-j)(z+j)}{z^3} = 1 - z^{-1} + z^{-2} - z^{-3}.$$

Correspondingly, by inverse z-transform, $h[n] = [1, -1, 1, -1]$.
(ii) The block diagram for this FIR filter is:



(iii)



(iv) Consider each term in $x[n]$ in turn.
The system $H(z)$ has zero gain at zero frequency hence the constant term is $3.7 \times 0 = 0$.
The system $H(z)$ has zero gain at frequency $\Omega = 2\pi/4$ hence the term in $\cos(n\pi/2)$ is also 0.
The system $H(z)$ has nonzero gain at frequency $\Omega = 2\pi/8$ given by

$$|H(e^{j\pi/4})| = |1 - \cos(\pi/4) + j\sin(\pi/4) - j + \cos(\pi/4) + j\sin(\pi/4)|$$
$$\text{since } \cos(3\pi/4) = -\cos(\pi/4) \text{ and } \sin(3\pi/4) = \sin(\pi/4)$$
$$\Longrightarrow |H(e^{j\pi/4})| = |1 - j(1 - 2\sin(\pi/4))| = 1.082.$$

The system $H(z)$ has phase at frequency $\Omega = 2\pi/8$ given by

$$\angle H(e^{j\pi/4}) = \arctan\left(\frac{-2\sin(\pi/4)}{1}\right) = 0.393.$$

Finally the gain and phase applied to the term at $\Omega = 2\pi/8$ gives

$$y[n] = -2.164\cos(n\pi/4 + 0.393).$$